Probability Exam 1, Timothy Pollard, RIN:662084381

Counting Methods

	W/ Ordering	W/o Ordering
W/ Replacement	n^k	$\binom{n+k-1}{k}$
W/o Replacement	$\frac{n!}{(n-k)!}$	$\binom{n}{k} = \frac{n!}{(n-k)!k!}$

Set Theory

Union: $A \cup B$
Intersection: $A \cap B$
Complement: A^c
Difference: $A - B = A \cap B^c$
Disjoint(Mutually Exclusive): $A \cap B = \emptyset$
DeMorgan's Law: $(A \cup B)^c = A^c \cap B^c$
Collectively Exhaustive: $\bigcup_{i=1}^n A_i = S$

Definitions

Experiment: Procedure, Observation, Model
Sample Space: set of all possible outcomes, S
Event: subset of sample space, $E_i \subseteq S$
Event Space: $\bigcup_{i=1}^N E_i = S, E_i \cap E_j = \emptyset$
Outcome: any possible result of an experiment, $s \in S$
Probability likelihood of an event A: $P(A)$

Axioms of Probability

1. For any event A, $P(A) \geq 0$
2. $P(S) = 1$
3. For any countable sequence of disjoint events $A_1, A_2, \ldots$, $P(\bigcup_{i=1}^\infty A_i) = \sum_{i=1}^\infty P(A_i)$

Bayes Rule & Axioms of Conditional Probability

Bayes Rule: $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$
Conditional Probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$
$P(A|B)P(B) = P(B|A)P(A)$

1. $P(A|B) \geq 0$
2. $P(B|B) = 1$
3. If $A = \bigcup_{i=1}^N A_i$ with $A_i \cap A_j = \emptyset$ then $P(A|B) = \sum_{i=1}^N P(A_i|B)$

Theroms

For any $A, B \subseteq S$:

1. $P(A^c) = 1 - P(A)$
2. $P(A) \leq 1$
3. $P(\emptyset) = 0$
4. If $B = \bigcup_{i=1}^n B_i$ and $B_i \cap B_j = \emptyset$ and $i \neq j$, Then $P(B) = \sum_{i=1}^n P(A_i)$
5. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
6. If $A \subseteq B$ then $P(A) \leq P(B)$
7. For any event $A \subseteq B$ and event space $\{B_1, B_2, \ldots, B_n\}$ then $P(A) = \sum_{i=1}^n P(A \cap B_i)$
8. If a sample space has outcomes that are equally likely, then for any event A, $P(A) = \frac{|A|}{|S|}$

Laws of Total Probability

1. Let $B_1, B_2 \ldots B_i$ be collectively exhaustive and mutually exclusive then for any event A, $A = \bigcup_{i=1}^N A \cap B_i$
2. $P(A) = \sum_{i=1}^N P(A \cap B_i) = \sum_{i=1}^N P(A|B_i)P(B_i)$

Independence

Independence: $P(A)P(B) = P(A \cap B)$
Conditional: $P(A|B) = P(A)$
Generalized: $P(\bigcap_{i=1}^N A_i) = \prod_{i=1}^N A_i$

Expected Value & Variance

$E[X] = \mu_X = \sum_{x \in S_X} x P_X(x) = \int_{-\infty}^{\infty} x f_X(x) dx$
Derived random variable $Y = g(X)$:
$E[Y] = \mu_Y = \sum_{x \in S_X} g(x) P_X(x) = \int_{-\infty}^{\infty} g(x) f_X(x) dx$
$Y = aX + b, a, b \in \mathbb{R}$ Then, $E[Y] = aE[X] + b$
$Y = X - \mu_X$ Then, $E[Y] = E[X - \mu_X] = 0$
Variance:
$\text{Var}(X) = \sum_{x \in S_X} (x - \mu_X)^2 P_X(x)$
$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx$
$\text{Var}(X) = \sigma_X^2 = E[X^2] - (E[X])^2 \quad \sigma_X = \sqrt{\text{Var}(X)}$
If Y = X+b, $\text{Var}(Y) = \text{Var}(X)$ Var(Const) = 0
If Y=aX, $\text{Var}(Y) = a^2 \text{Var}(X)$

Random Variables & PMF

Discrete, finite and continuous(PDF)
Random variables map Sample Space to $\mathbb{R}$:
$S = \{\zeta\}, S_X = X(\zeta) \in \mathbb{R} | \zeta \in S, S_X \subseteq \mathbb{R}$
Equivalent Events: $A = \{\zeta | X(\zeta) \in B\}$
For a random variable X w/ $P_X(x)$ and range S_X

1. For any x $P_X(x) \geq 0$
2. $\sum_{x \in S_X} P_X(x) = 1$
3. For any event $B \subset S_X, P(B) = \sum_{x \in S_B} P_X(x)$

Common Discrete R.Vs:

Name	**PMF** $P_X(x)$	**E[x]**
Bernoulli	$\begin{cases} 1-p, & x=0 \\ p, & x=1 \\ 0, & \text{otherwise} \end{cases}$	p
Geometric	$\begin{cases} p(1-p)^{x-1}, & x=1,2,\ldots \\ 0, & \text{otherwise} \end{cases}$	$\frac{1}{p}$
Binomial	$\begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x=0,1,\ldots,n \\ 0, & \text{otherwise} \end{cases}$	np
Pascal	$\begin{cases} \binom{x-1}{k-1} p^k (1-p)^{x-k}, & x=k,k+1,\ldots \\ 0, & \text{otherwise} \end{cases}$	$\frac{k}{p}$
Uniform	$\begin{cases} \frac{1}{l-k+1}, & x=k,k+1,\ldots,l \\ 0, & \text{otherwise} \end{cases}$	$\frac{k+l}{2}$
Poisson	$\begin{cases} \frac{\alpha^x e^{-\alpha}}{x!}, & x=0,1,2,\ldots \\ 0, & \text{otherwise} \end{cases}$	α

PDF: $\frac{dF_X(x)}{dx} = f_X(x)$

CMF/CDF

$$F_X(x) = P(X \leq x) = \sum_{y=-\infty}^{x} P_X(x)$$

$\lim_{x \to -\infty} F_X(x) = 0, \quad \lim_{x \to +\infty} F_X(x) = 1$
$F_X(x) = F_X(x_i)$ for all $x_i \leq x < x_{i+1}$
$F_X(b) - F_X(a) = P(a < x \leq b)$

$$F_X(x) = \int_{-\infty}^{x} f_X(u) du$$

Variance of Common R.V.

R.V.	**Parameters**	**Var(X)**
Bernoulli	$p \in [0,1]$	$p(1-p)$
Geometric	$p \in (0,1]$	$\frac{1-p}{p^2}$
Binomial	$n \in \mathbb{N}, p \in [0,1]$	$np(1-p)$
Pascal	$k \in \mathbb{N}, p \in (0,1]$	$\frac{k(1-p)}{p^2}$
Poisson	$\alpha > 0$	α
Uniform	k, l	$\frac{(l-k)(l-k+2)}{12}$